IoT: Architecture and Protocol

Đánh giá học phần

Thành phần đánh giá	Trọng số (%)	Hình thức đánh giá	Trọng số con (%)
(1)	(2)	(3)	(4)
A1. Đánh giá quá trình	20%	A1.1. Chuyên cần	40%
		A1.2. Bài tập	60%
A2. Đánh giá giữa kỳ	20%	A2.1. Tự luận	100%
A3. Đánh giá cuối kỳ	60%	A3.1. Đồ án	100%

Chapter 1: Overview

What is IoT?

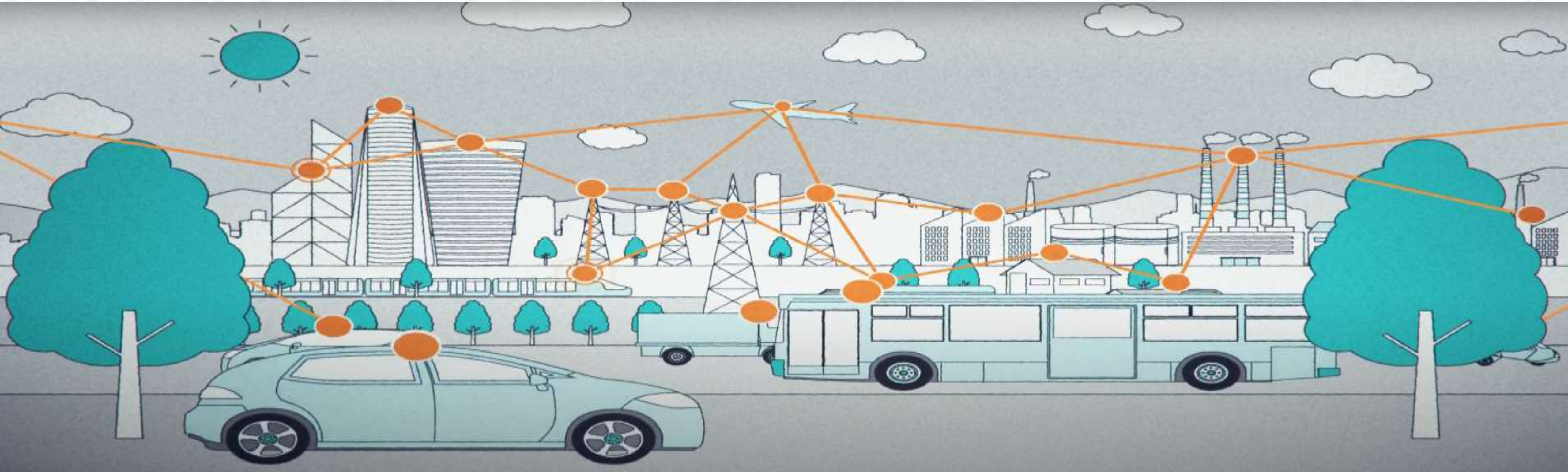
- Connecting everyday things embeded with electronics, software and sensor to the internet enabling them to collect and exchange data

Why do we need IoT?

- Hospital assistance
- Traffic control and monitor
- Home electrical device
- Human health
- ……

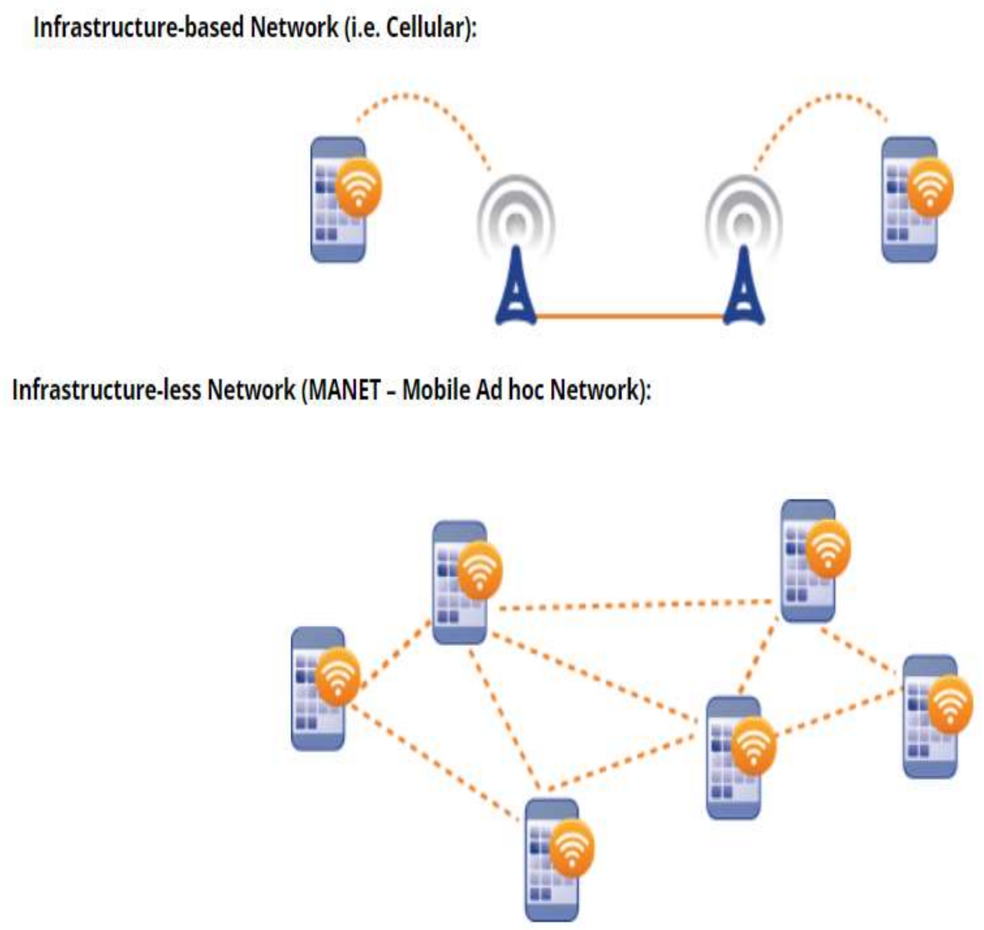
Wireless Ad-hoc and Sensor Networks

- Wireless sensor network (WSNs)
  - Wireless : Mobility support and ease of system deployment
  - Sensor : capability of sensing technology to provide the means to perceive and interact
  - Network : the possibility of building systems whose functional capabilities



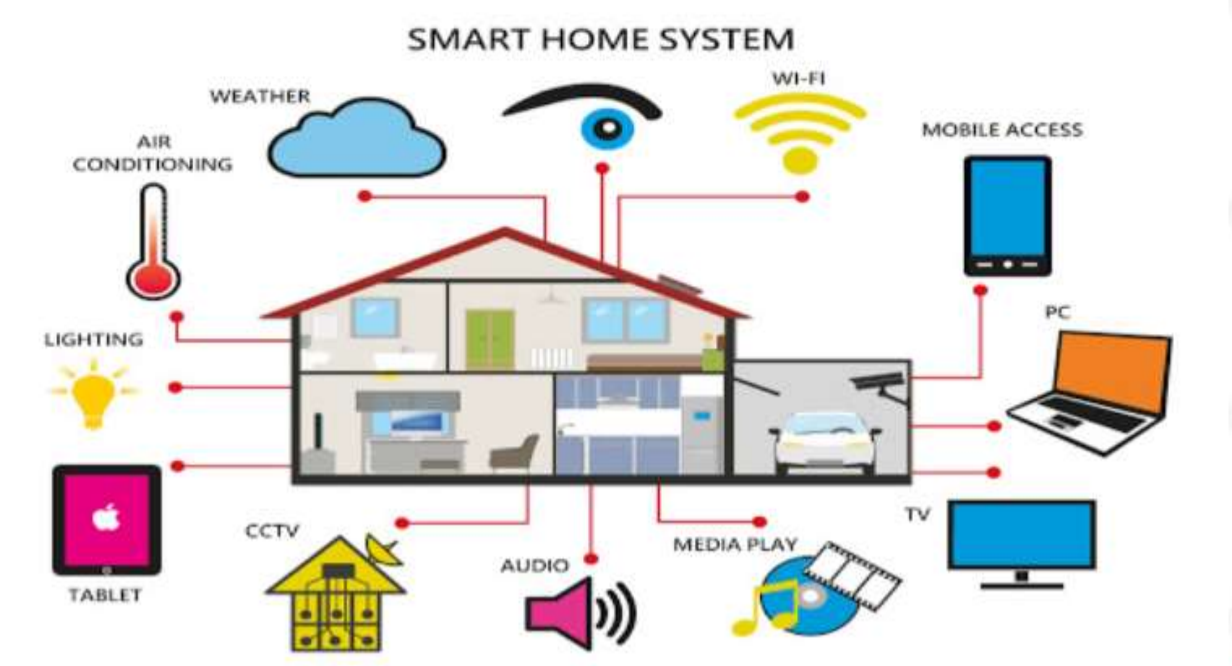
Wireless Ad-hoc and Sensor Networks

- Requirement of IoT network model
  - Self-configuration and self-organization in infrastructure-less systems
  - Support for cooperative operations in systems with heterogenous members
  - Multi-hop peer-to-peer communication among network nodes, with effective routing protocols
  - Network self-healing behavior providing a sufficient degree of robustness and reliability
  - Seamless mobility management and support of dynamic network topologies



IoT-Enable Application

- **Home and Building Automation (smart home)**
  - Building automation in a private home
  - Convenience and security

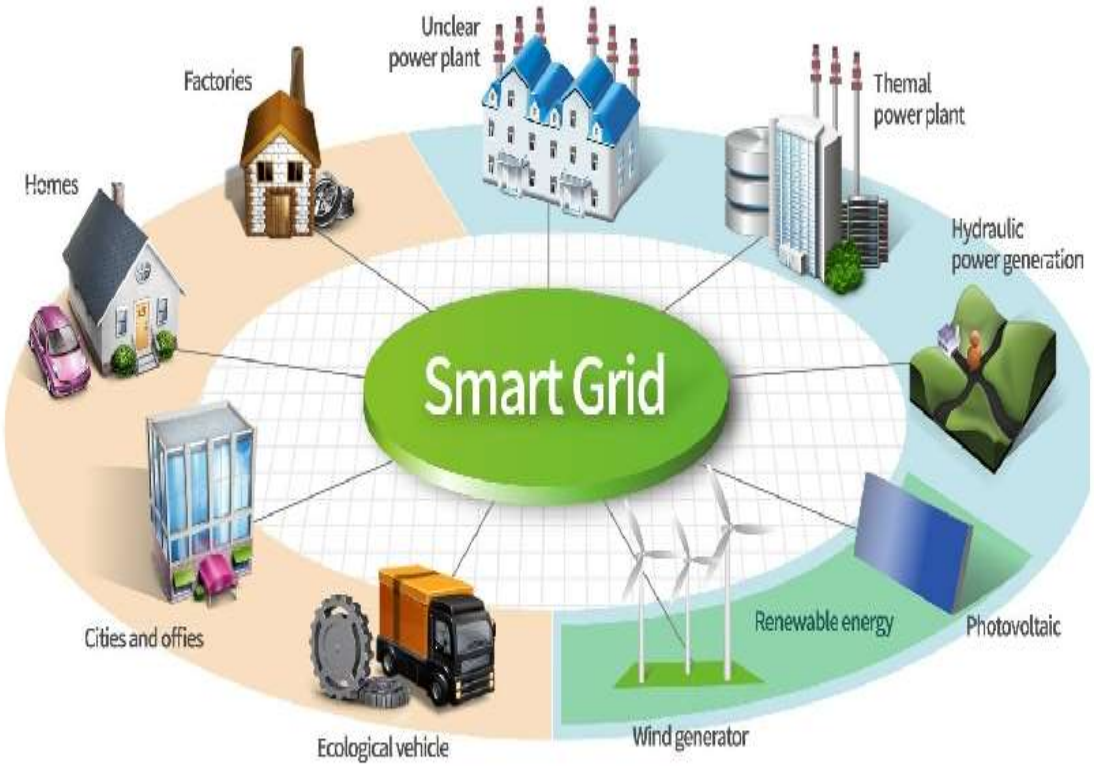


IoT-Enable Application

- Smart city
  - Complex ecosystems
  - Collecting data on the city status and disseminating them to citizens

IoT-Enable Application

- Smart Grids
  - A variety of operational systems, including smart meters, smart appliances, renewable energy resources, and energy-efficient resources
  - Power line communication
    - Use of existing electrical cables to transport data
    - Enabler for sensing, control, and automation in large systems spread over relatively wide areas



IoT-Enable Application

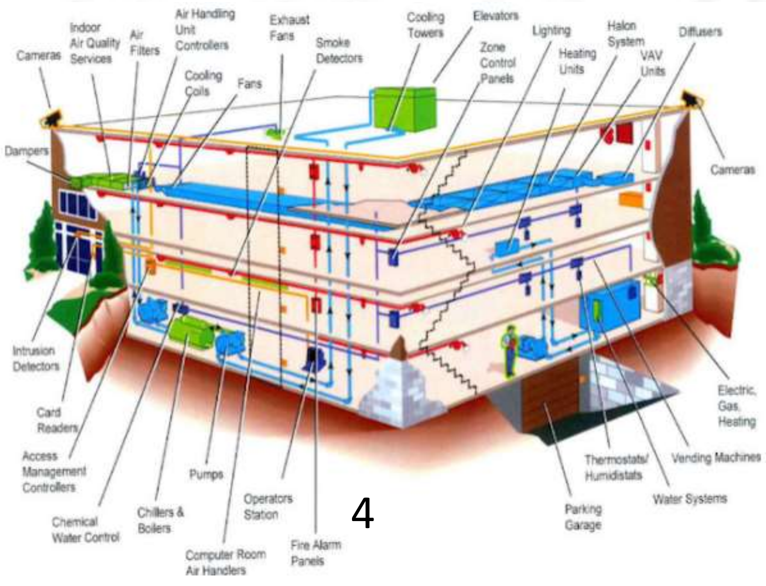
- **Industrial IoT** (IIoT)
  - Manufacturing, logistics, oil and gas, transportation, energy/utilities, mining and metals, aviation …..
    - *real-time* responses are critical
    - Be inability to share information between devices
- 3 IIoT capabilities must be mastered:
  - *sensor-driven computing*
    - converting sensed data into insights operators and systems can act on
  - *industrial analytics*
    - turning data from sensors and other sources into actionable insights
  - *intelligent machine applications*
    - integrating sensing devices and intelligent components into machines

IoT-Enable Application

- Smart farming (Agriculture 4.0)
  - Lead to timely, cost-effective, efficient farming systems while also tackling environmental impacts
  - Quickly respond to any significant change in weather, humidity and air quality, as well as the health of each crop or soil in the field
- Challenges
  - Farm size
  - More frequent use of vehicles
  - Excessive data and highly variable conditions

CLASSIFICATION

1. Environmental monitoring
2. Precision agriculture
3. Machine and process control
4. Facility automation
5. Traceability systems



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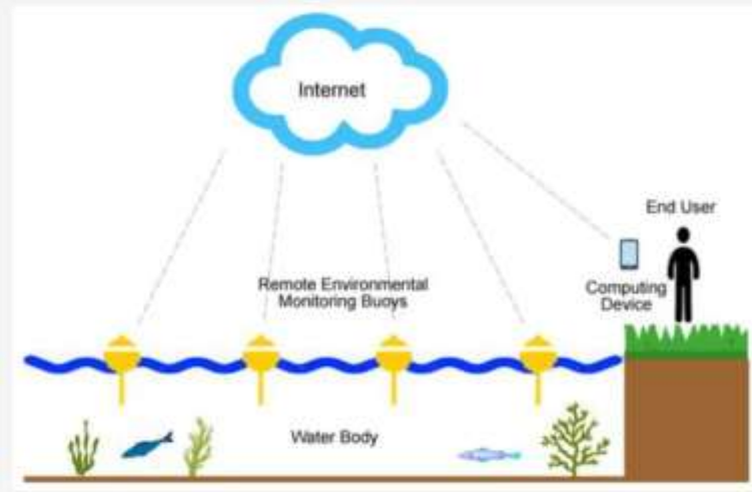

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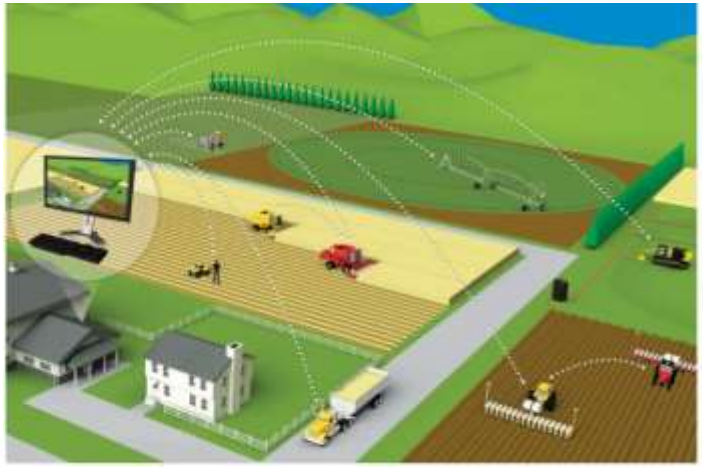

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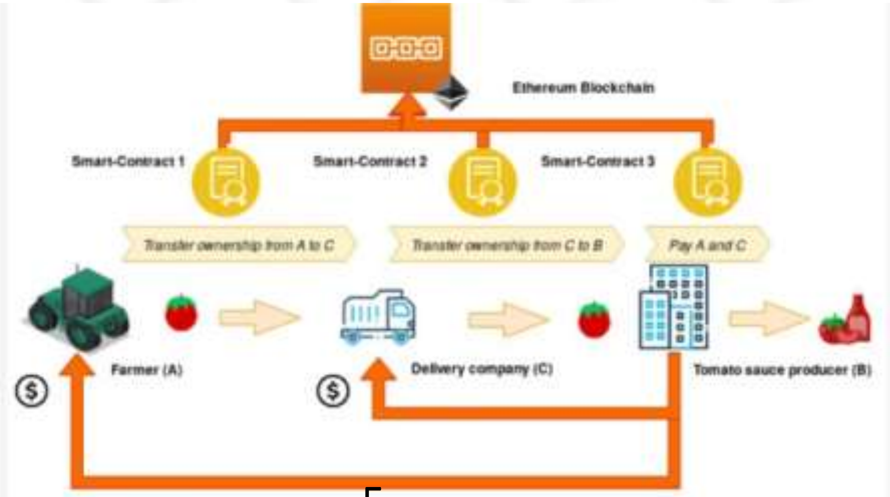

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